2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSE-1004

(Image Processing)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1, 2,** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

   (a) Write the definition of Gaussian notch filter transfer function.

   (b) What is image contrast ratio?

   (c) What is the significance of Histogram Equalization?

   (d) Write the difference between CMY and CMYK model.

   (e) Why is Intensity Slicing done?

   (f) Write the definition of Point Spreading Function.

   (g) Why the Median filter is effective in eliminating Paper-Salt noise?

2. Answer ***any five*** questions : 4×5

   (a) Mathematically explain the Color Transformation of HSI and RGB.

   (b) Explain Contrast Stretching for digital image processing.

   (c) Explain the Histogram Matching technique for image enhancement.

   (d) Develop an algorithm for converting an one-pixel-thick 8-path to a 4-path.

   (e) Explain Log Transformation for digital image.

   (f) Write the advantages and disadvantages of Gamma Transformation.

   (g) Explain System Transfer function in the frequency domain with an example.

3. (a) What is median filter?

   (b) How the median filter works?

   (c) Write the advantages and disadvantages of it in image processing. 2+4+4

4. (a) Differentiate between brightness and hue.

(b) Derive the CMY transformations to generate the complement of a color image.

(c) What is the color slicing? — Explain with example. 3+4+3

5. (a) What is meant by fuzzification?

(b) Write a simple fuzzy based image enhancement algorithm using fuzzy normalization technique.

(c) Why fuzzy theory is important for image processing? 2+5+3

6. (a) What is meant by masking?

(b) Explain different types of image noises.

(c) Explain removal of image noises by different types of linear spatial domain filter. 2+3+5

7. (a) What is maximum and minimum filter?

(b) Write the different blurring techniques.

(c) Write the advantages and disadvantages of bi-lateral filter. 2+4+4

8. (a) Compare brightness with contrast.

(b) Explain different types of image enhancement technique based on image histograms.

(c) Write applications of image enhancement in image processing. 3+4+3